Title

Restaurant Recommendation System using Machine Learning

1. Introduction

In today's digital world, users often face difficulty in choosing restaurants based on their preferences. This project aims to build a Restaurant Recommendation System that suggests restaurants based on user inputs such as cuisine, city, and budget.

The system uses machine learning techniques to filter and recommend the best restaurants.

2. Objective

To build a recommendation system using restaurant dataset

To suggest top restaurants based on user preferences

To improve user decision-making using data

3. Dataset Description

The dataset contains information about restaurants such as:

Restaurant Name

City

Cuisines

Average Cost for Two

Aggregate Rating

Total Records: 9551

Total Features: 21

4. Methodology

*Step 1: Data Preprocessing

Removed unnecessary columns

Handled missing values

Selected important features

*Step 2: Feature Engineering

Combined cuisines and city

Used cost as filtering parameter

*Step 3: Model Used

Content-Based Recommendation System

TF-IDF Vectorization

Cosine Similarity

*Step 4: Recommendation Logic

Filter restaurants based on:

Cuisine

City

Budget (Average Cost)

Sort results based on rating

5. Implementation Tools

Python

Google Colab

Pandas

Scikit-learn

6. Result

The system successfully recommends top restaurants based on user input.

🔷 Sample Input:

Cuisine: North Indian

City: Delhi

Budget: 1000

🔷 Output:

List of top 5 restaurants with highest ratings

👉 The model provides accurate and relevant recommendations.

7. Advantages

Easy to use

Fast recommendations

Personalized results

Helps users save time

8. Limitations

Depends on dataset quality

No real-time data

Limited personalization

9. Future Scope

Add user login system

Use real-time APIs (like Zomato/Swiggy)

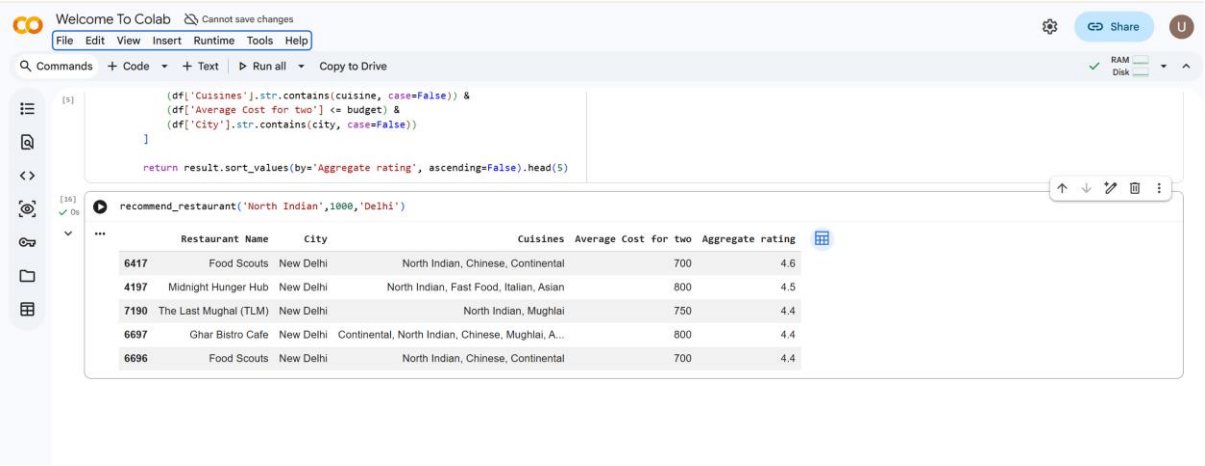
Build web app or mobile app

Add reviews-based sentiment analysis

10. Conclusion

The project demonstrates how machine learning can be used to build an efficient recommendation system. It helps users find suitable restaurants quickly based on their preferences.

Output:



	Restaurant Name	City	Cuisines	Average Cost for two	Aggregate rating
6417	Food Scouts	New Delhi	North Indian, Chinese, Continental	700	4.6
4197	Midnight Hunger Hub	New Delhi	North Indian, Fast Food, Italian, Asian	800	4.5
7190	The Last Mughal (TLM)	New Delhi	North Indian, Mughlai	750	4.4
6697	Ghar Bistro Cafe	New Delhi	Continental, North Indian, Chinese, Mughlai, A...	800	4.4
6696	Food Scouts	New Delhi	North Indian, Chinese, Continental	700	4.4